Experiment No. 07

Aim: 3D Plotting.

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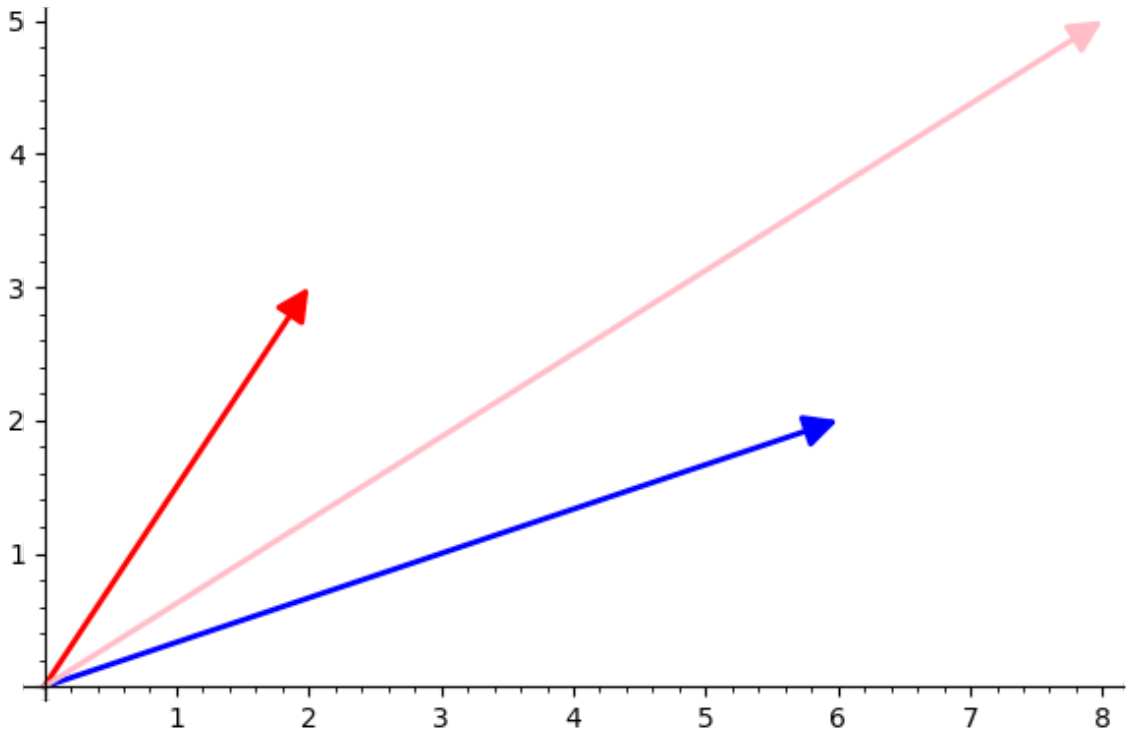
Section_batch_RollNo:A7_B3_59.

In [148]:
```
arrow3d((1,1,1),(2,2,2),4,color="red")
```

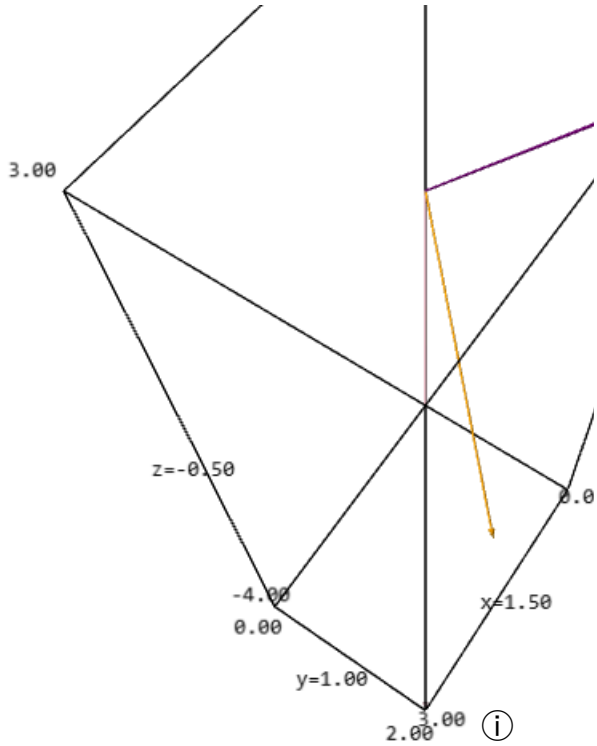
Out[148]:

In [149]:
```
a=arrow3d((0,0,0),(1,1,1),6,color="blue")
b=arrow3d((0,0,0),(1,2,3),6,color="red")
c=arrow3d((0,0,0),(1,-1,1),6,color="green")
show(a+b+c)
```

In [131]:
```
a=vector([2,3])
b=vector([6,2])
c=a+b
p1=plot(a,color="red")
p2=plot(b,color="blue")
p3=plot(c,color="pink")
show(p1+p2+p3)
```

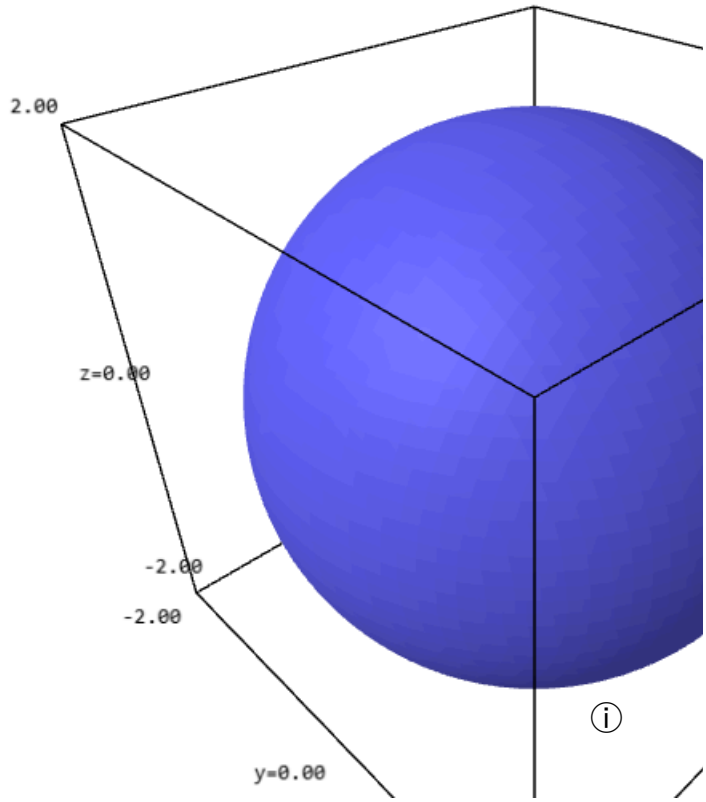


In [132]:
```
a=vector([1,2,3])
b=vector([2,2,-1])
c=vector([3,2,-4])
p1=plot(a,color="purple")
p2=plot(b,color="orange")
p3=plot(c,color="pink")
show(p1+p2+p3)
```



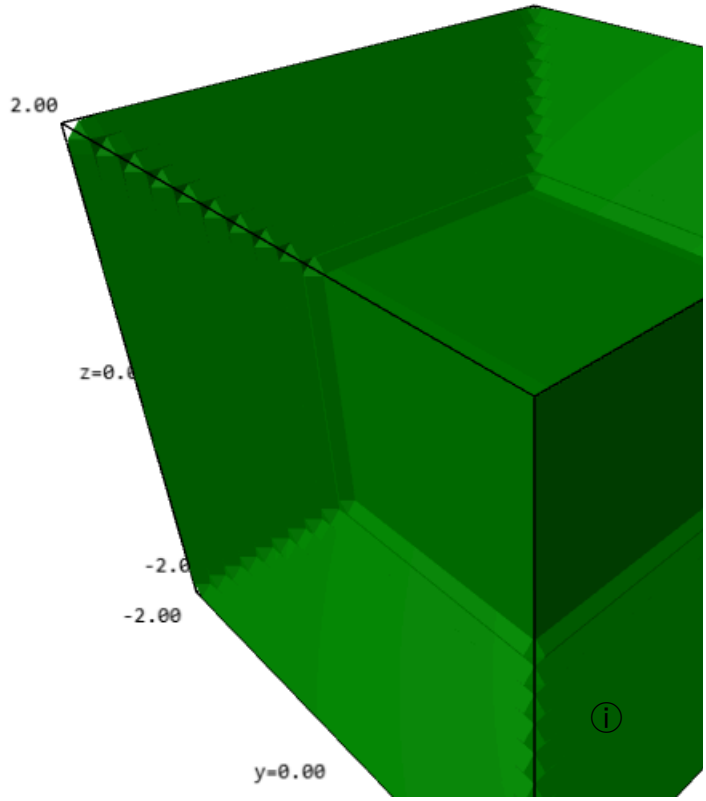
In [133]:
```
var('x,y,z')
n=2
f(x,y,z)=x^n+y^n+z^n-4
implicit_plot3d(f,(x,-2,2),(y,-2,2),(z,-2,2))
```

Out[133]:



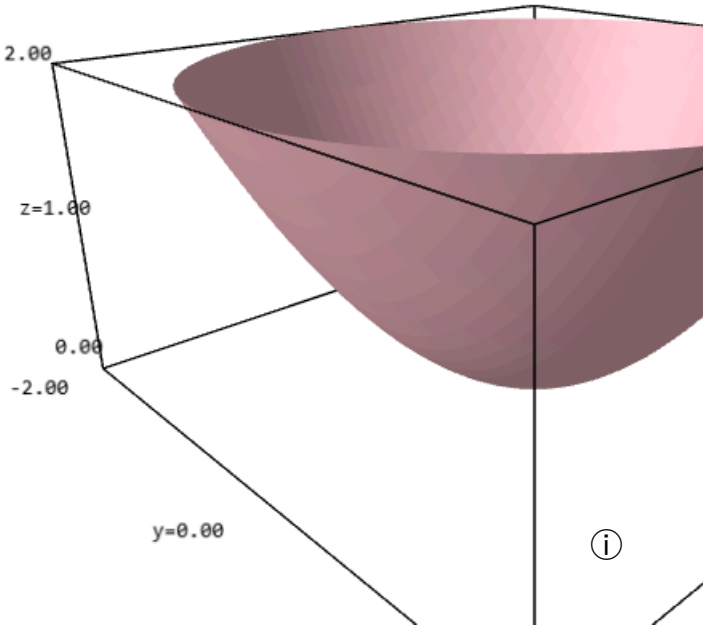
In [134]:
```
var('x,y,z')
n=45
f(x,y,z)=x^n+y^n+z^n-4
implicit_plot3d(f,(x,-2,2),(y,-2,2),(z,-2,2),color="green")
```

Out[134]:



In [135]:
```
var('x,y,z')
f(x,y,z)=x^2+y^2-2*z
implicit_plot3d(f,(x,-2,2),(y,-2,2),(z,0,2),color="pink")
```

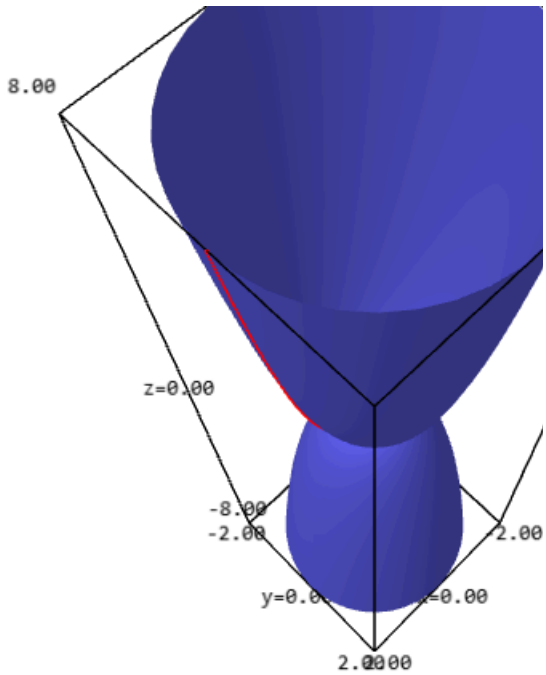
Out[135]:



Surface of revolution

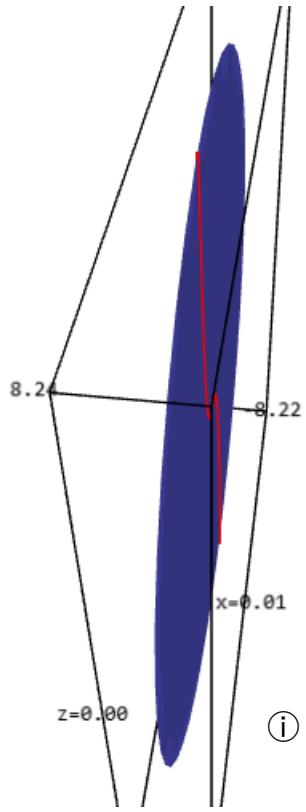
In [136]:
```
revolution_plot3d(x^3,(x,-2,2),parallel_axis='z',show_curve=True)
```

Out[136]:



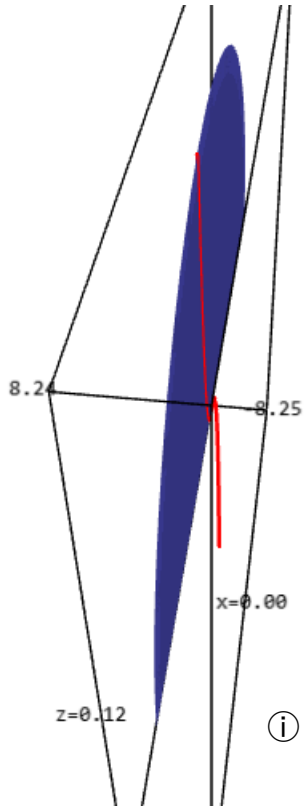
In [137]:
```
revolution_plot3d(x^3,(x,-2,2),parallel_axis='y',show_curve=True)
```

Out[137]:



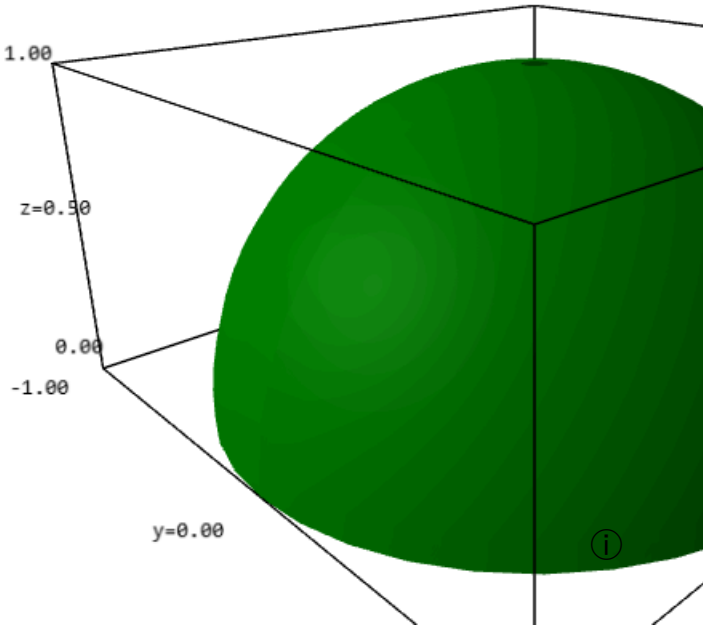
In [138]:
```
revolution_plot3d(x^3,(x,-2,2),(0,pi),parallel_axis='y',show_curve=True)
```

Out[138]:



In [139]:
```
var('t')
revolution_plot3d((cos(t),sin(t)),(t,0,pi),parallel_axis='z',color="green")
```

Out[139]:



Exercise

1) Find the surface of revolution of y=x^3/(x+1)around z axis through an angle 0 to 2 pi and through and x- axis through an angle 0 to 2 pi and 0 to pi

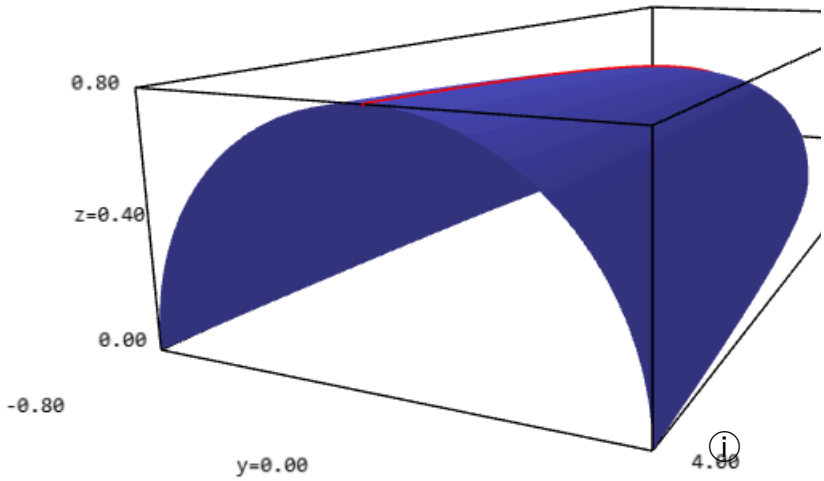
In [140]:
```
var('x')

# Around Z-axis, 0 to 2π
revolution_plot3d(x/(x+1), (x,0,4), (0,2*pi), parallel_axis='z', show_curve
=True)

# Around X-axis, 0 to 2π
revolution_plot3d(x/(x+1), (x,0,4), (0,2*pi), parallel_axis='x', show_curve
=True)

# Around X-axis, 0 to π
revolution_plot3d(x/(x+1), (x,0,4), (0,pi), parallel_axis='x', show_curve=T
rue)
```

Out[140]:



Find the surface of revolution of around z axis through an angle to and through

and x- axis through an angle to and to

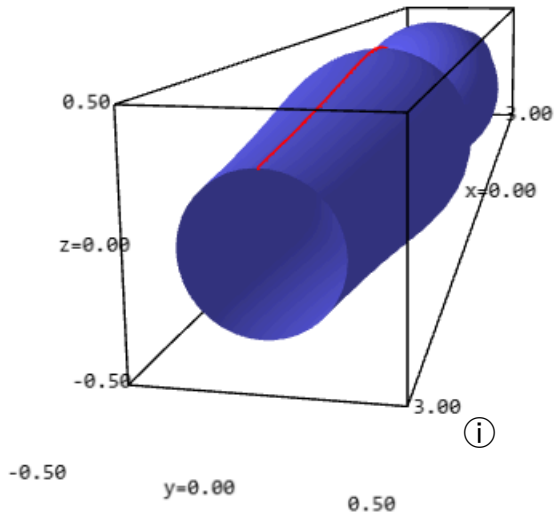
y = x/(+ 1) x2 0 π 0 2π 0 2π

In [141]:
```
var('x')

# Around Z-axis, 0 to π
revolution_plot3d(x/(x^2+1), (x,-3,3), (0,pi), parallel_axis='z', show_curv
e=True)

# Around X-axis, 0 to 2π
revolution_plot3d(x/(x^2+1), (x,-3,3), (0,2*pi), parallel_axis='x', show_cu
rve=True)
```

Out[141]:



3) plot the graph of the following curves \ (a) within

the range \ (b) within the range \ (c) within

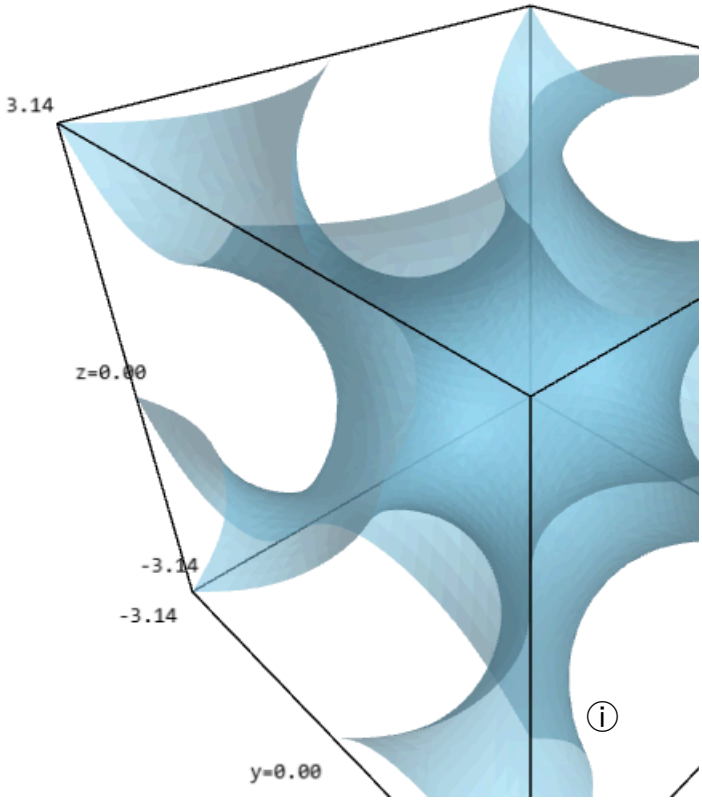
the range for different values of

sin(x) cos(y) + sin(y) cos(z) + sin(z) cos(x) = 0

x ∈ [−π, π], y ∈ [−π, π], z ∈ [−π, π]. x − 3x − z = 0 3 y2 x ∈ [−2, 2], y ∈ [−2, 2], z ∈ [−2, 2]. (R − x + + − = 0, R > r 2 y √− −−−−−2)2 z2 r2 x ∈ [−3, 3], y ∈ [−3, 3], z ∈ [−2, 2]. R, r

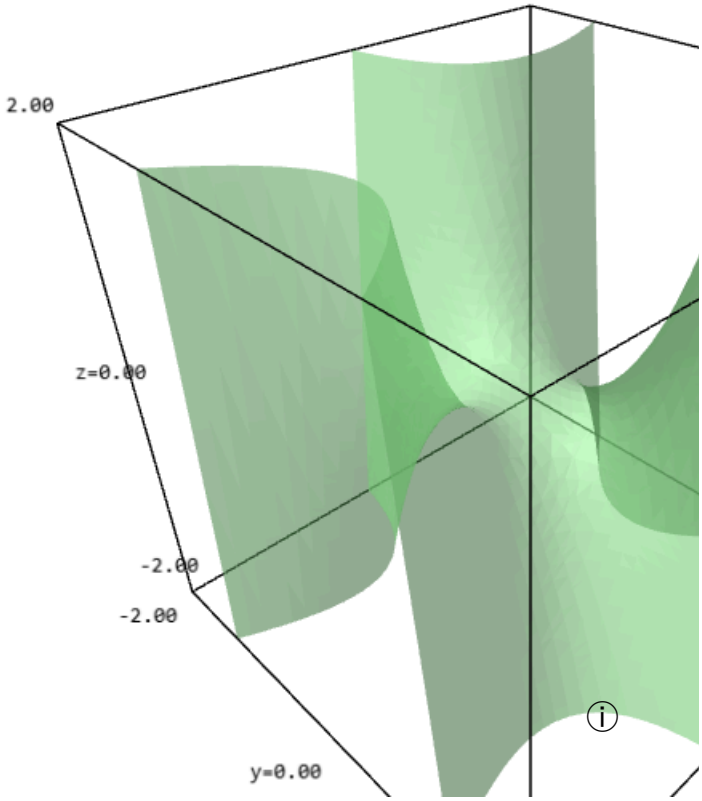
In [142]:
```
# (a)
var('x,y,z')
f(x,y,z) = sin(x)*cos(y) + sin(y)*cos(z) + sin(z)*cos(x)
implicit_plot3d(f, (x,-pi,pi), (y,-pi,pi), (z,-pi,pi), opacity=0.6, color
='skyblue')
```

Out[142]:



In [143]:
```
#(b)
var('x,y,z')
f(x,y,z) = x^3 - 3*x*y^2 - z
implicit_plot3d(f, (x,-2,2), (y,-2,2), (z,-2,2), opacity=0.6, color='lightg
reen')
```

Out[143]:

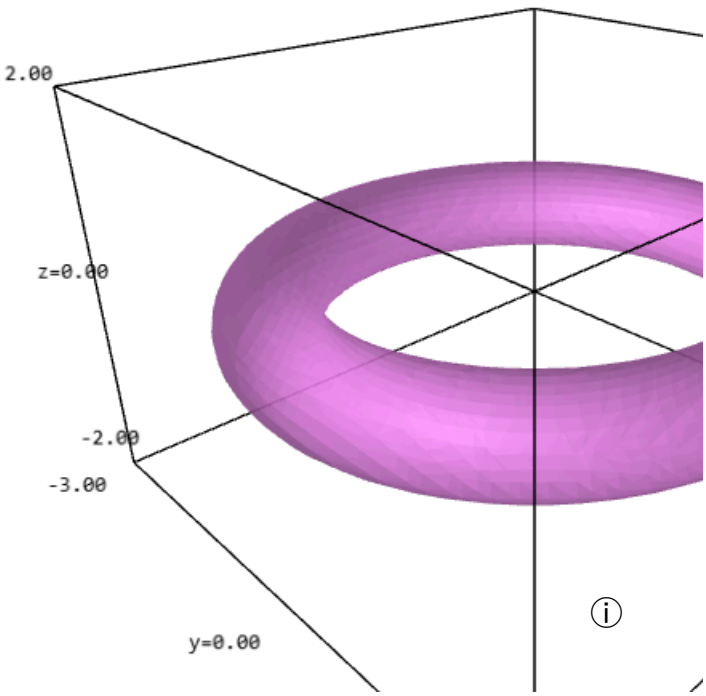


In [144]:
```
#(c)
var('x,y,z')

# Try R=2, r=1
R=2; r=1
f(x,y,z) = (R - sqrt(x^2+y^2))^2 + z^2 - r^2
implicit_plot3d(f, (x,-3,3), (y,-3,3), (z,-2,2), color='orange', opacity=0.
7)

# Try R=2.5, r=0.5
R=2.5; r=0.5
f(x,y,z) = (R - sqrt(x^2+y^2))^2 + z^2 - r^2
implicit_plot3d(f, (x,-3,3), (y,-3,3), (z,-2,2), color='violet', opacity=0.
7)
```

Out[144]:



4) Find the revolution plot along axes of the following curves with 360 degree rotation\ (a)

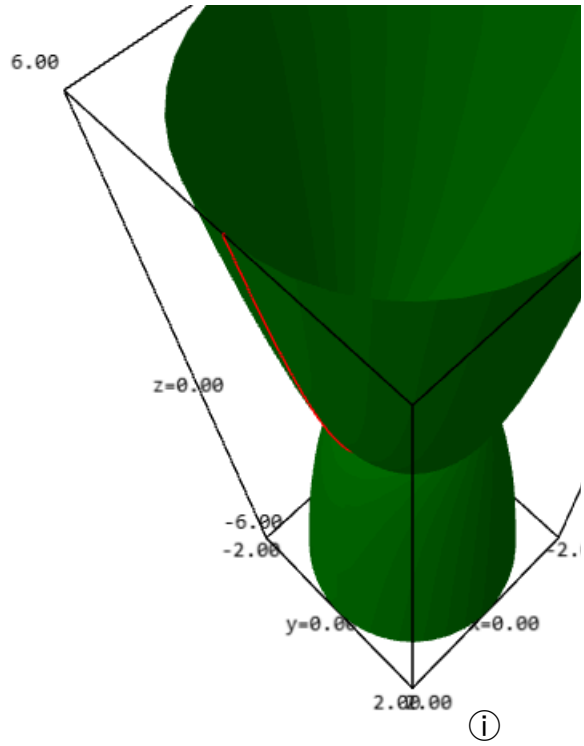
(b) (c) x, y, z

y = − xx3 y = e

√ x)(y = ln(x + 1)

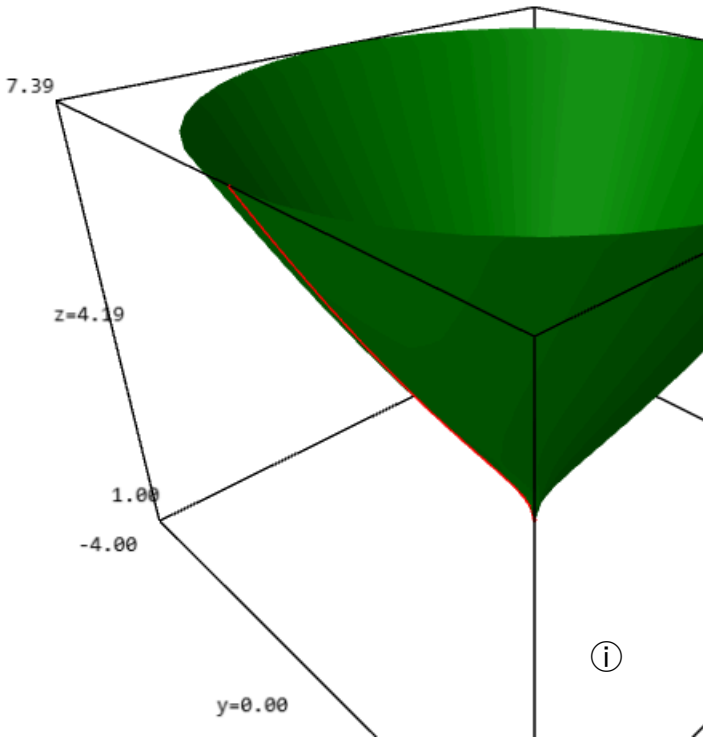
In [145]:
```
#(a)
var('x')
revolution_plot3d(x^3-x, (x,-2,2), (0,2*pi), parallel_axis='x', show_curve=
True, color='red')
revolution_plot3d(x^3-x, (x,-2,2), (0,2*pi), parallel_axis='y', show_curve=
True, color='blue')
revolution_plot3d(x^3-x, (x,-2,2), (0,2*pi), parallel_axis='z', show_curve=
True, color='green')
```

Out[145]:



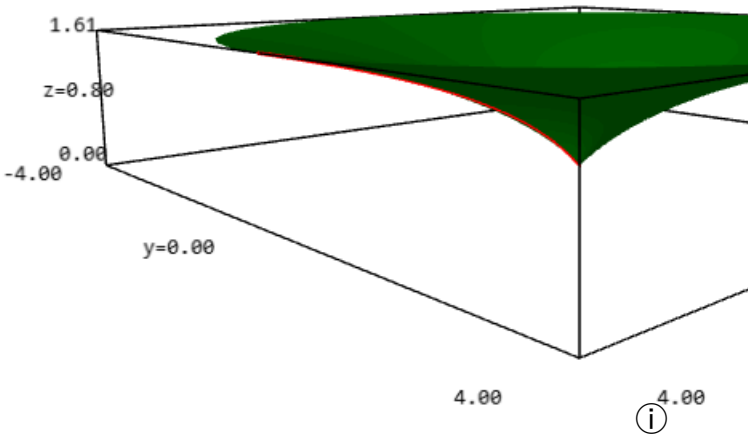
In [146]:
```
#(b)
var('x')
revolution_plot3d(e^sqrt(x), (x,0,4), (0,2*pi), parallel_axis='x', show_cur
ve=True, color='red')
revolution_plot3d(e^sqrt(x), (x,0,4), (0,2*pi), parallel_axis='y', show_cur
ve=True, color='blue')
revolution_plot3d(e^sqrt(x), (x,0,4), (0,2*pi), parallel_axis='z', show_cur
ve=True, color='green')
```

Out[146]:



In [147]:
```
#(c)
var('x')
revolution_plot3d(ln(x+1), (x,0,4), (0,2*pi), parallel_axis='x', show_curve
=True, color='red')
revolution_plot3d(ln(x+1), (x,0,4), (0,2*pi), parallel_axis='y', show_curve
=True, color='blue')
revolution_plot3d(ln(x+1), (x,0,4), (0,2*pi), parallel_axis='z', show_curve
=True, color='green')
```

Out[147]:



Learning:(Write in details about your Learnings from the activity):

In this experiment, I learned how to use SageMath as a computational tool to visualize mathematical concepts in three dimensions, which helped me understand topics from Calculus and Linear Algebra in a much more intuitive way.

1. 3D Arrow and Vector Plotting: I learned how to draw vectors and arrows in 3D space using the arrow3d() and vector() commands. By specifying starting and ending points along with color and width, I could visually represent vectors and observe their direction and magnitude in 3D. I also learned how to add multiple vector plots together using the show() command, which helped me visualize vector addition geometrically — something that was previously only theoretical for me.
2. Implicit 3D Surface Plotting: I learned how to plot surfaces defined by implicit equations of the form f(x, y, z) = 0 using the implicit_plot3d() command. I observed that changing the value of the power n in equations like $x^n + y^n + z^n - 4 = 0$ produces very different surface shapes — for even values of n it gives a closed surface, while odd values give an open surface. I also learned to control visual properties like color and opacity to make graphs easier to interpret.
3. Surface of Revolution: This was the most interesting part of the experiment. I learned that when a 2D curve is rotated about an axis through a certain angle, it generates a 3D solid called a surface of revolution. Using the revolution_plot3d() command, I could revolve curves like $y = x^3$, $y = x^2$, or $y = \sqrt{x}$ around the x, y, or z axis. I also learned the difference between rotating through 0 to π (half rotation) and 0 to 2π (full rotation) — a full rotation gives a complete solid while a half rotation gives an open shell shape. This concept is very useful in real-world applications like designing containers, pipes, and machine parts.
4. Combining Multiple Plots: I learned how to combine two or more separate plots into a single visualization using the + operator between plot objects and then using show(). This was useful in comparing two surfaces simultaneously, like a cylinder and a paraboloid overlapping in the same 3D space.
5. Parametric and Implicit 2D Curves: I also learned how to plot parametric curves like x = cos(t), y = sin(t) and use them as input for revolution plots. Additionally, I explored implicit_plot() for 2D curves defined by equations that cannot be easily written as y = f(x).
6. Connecting Theory to Visualization: Overall, this experiment taught me that mathematical equations are not just abstract symbols — they represent actual geometric shapes and surfaces. Visualizing them using code gave me a much deeper understanding of concepts like vector spaces, surface equations, and solids of revolution that I had previously only studied theoretically in my Calculus and Linear Algebra course.

In [ ]: